

**THE OPEN UNIVERSITY OF SRI LANKA
FACULTY OF ENGINEERING TECHNOLOGY**

COMPUTER ENGINEERING

ACADEMIC YEAR :– 2025/26

EES5326: - Digital Electronic Design

LAB Sheet: LAB 1



During these two days you will be introduced to the digital electronic design environment. Major parts of it can be seen as VHDL modeling tools, different modeling strategies, and Field Programmable Gate Arrays (FPGAs). Also, you will design and implement some of your designs on a FPGA and test whether it is producing the expected output. Moreover, you will test different kind of digital components with FPGA.

You are expected to simulate, verify and implement your designs. You need to be aware of the steps carried out in order to implement a design on an FPGA, and also the resource usage of the FPGA.

Aim: This lab will prepare learners with essential skills and in HDL-based digital design and FPGA implementation, preparing them for advanced projects in digital electronic system design in hierarchical fashion.

Objectives:

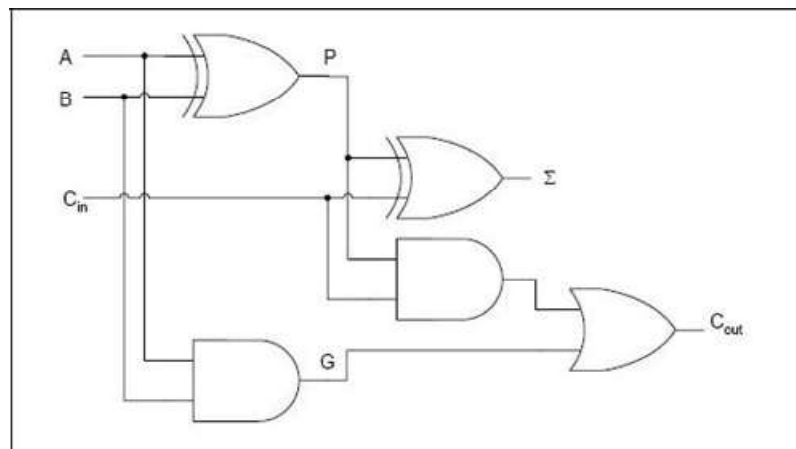
1. Design and model basic combinational and sequential logic circuits using VHDL.
2. Simulate and verify digital designs using appropriate tools to ensure correct functionality.
3. Implement and test VHDL designs on an FPGA development board, observing real-time behavior.
4. Understand the FPGA implementation flow, including synthesis, and bitstream generation.
5. Analyze FPGA resource usage (LUTs, flip-flops, memory blocks).

Note: Use of Generative AI will receive Zero marks.

You may complete up to section A during the Day 1.

Section A – Combinational Logic

- 1 Read the Anvyl FPGA Demonstration manual and follow the instructions. Test the LED demo, Counter demo, Keypad demo and Touch pad demo with Anvyl FPGA.
- 2 Design an INVERTER, AND gate, OR gate, XOR gate, Multiplexer and D-Flip-flop. Select the number of inputs according to your preference.
- 3 Using previous logic gates, design a unit shown in figure 01.



- 4 Draw a timing diagram for your proposed design
- 6 Model your ASIC in VHDL and simulate it. Implement and test your design with FPGA.

Section B

Sequential Logic

- 1 Design and simulate T-Flip-Flop using Behavioral or Structural or Dataflow modeling method.
- 2 Consider the circuit in Figure 2. It is a 4-bit synchronous counter which uses four T-type flip-flops. The counter increments its value on each positive edge of the clock if the Enable signal is asserted. The counter is reset to 0 by setting the Clear signal low. You are to implement the 4-bit counter given below.

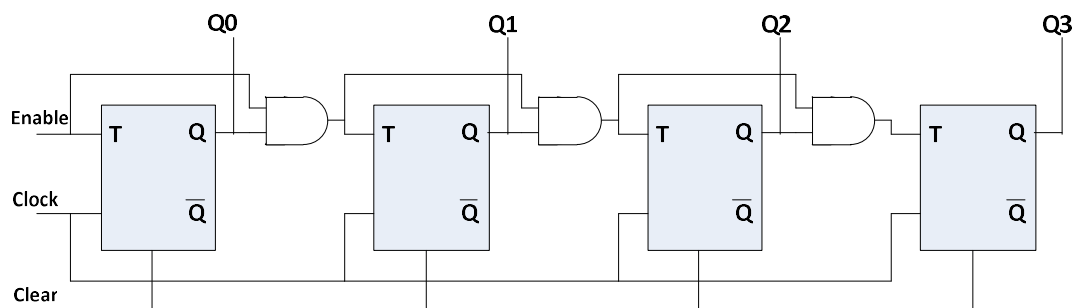


Figure 2: A 4-bit counter.

- 3 Write a VHDL code applying Structural model using T flip-flop as a component.
- 4 Draw the timing diagram manually. Simulate implementation and analyze using logic analyzer.

Hierarchical Design

- 1 You are expected to prepare your own hierarchical design for the following situation. a and b are eight-bit_buses. Identify the special functional units and build the structural model using already identified functional units.

```

If a<b then
    d= a + b
    c= a * b
else
    d= b + 1
    c= a - 1
end if

```

- a. Draw a timing diagram for your proposed design
- b. Model your hierarchical design in VHDL and simulate it. You have to create all required components (e.i Invertors and Multiplexers, etc) and you should use previously design full adder.
- c. Implement the given hierarchical design on FPGA development board and verify the results.